

SEMICONDUCTOR DEFECT INSPECTION REPORT

Enterprise AI-Powered Quality Inspection System | MIPHI Program | CUBE AI Solutions

INSPECTION SUMMARY

FAIL — Scratch defect detected

PREDICTION DETAILS

Field	Value
Filename	wafer_sample_001.png
Defect Classification	Scratch
Confidence Score	92.30%
Severity Level	HIGH
Inspection Result	FAIL
Inference Time	0.187 seconds
AI Model	YOLOv8n-cls
Model Accuracy	92.05% (on 9,370 test images)
Inspection Timestamp	2026-07-04 22:19:36

DEFECT DESCRIPTION

Linear scratch detected across wafer surface — likely physical handling damage during transport or processing.

DEFECT SEVERITY REFERENCE

Defect Type	Severity	Action Required
none	PASS	No action needed — wafer approved
Random	LOW	Monitor — likely equipment noise

Loc	MEDIUM	Investigate contamination source
Edge-Loc	MEDIUM	Check edge handling process
Center	HIGH	Inspect center processing equipment
Donut	HIGH	Check chemical distribution system
Edge-Ring	HIGH	Inspect edge seal and handling
Scratch	HIGH	Review physical handling procedures
Near-full	CRITICAL	Halt production — major equipment failure